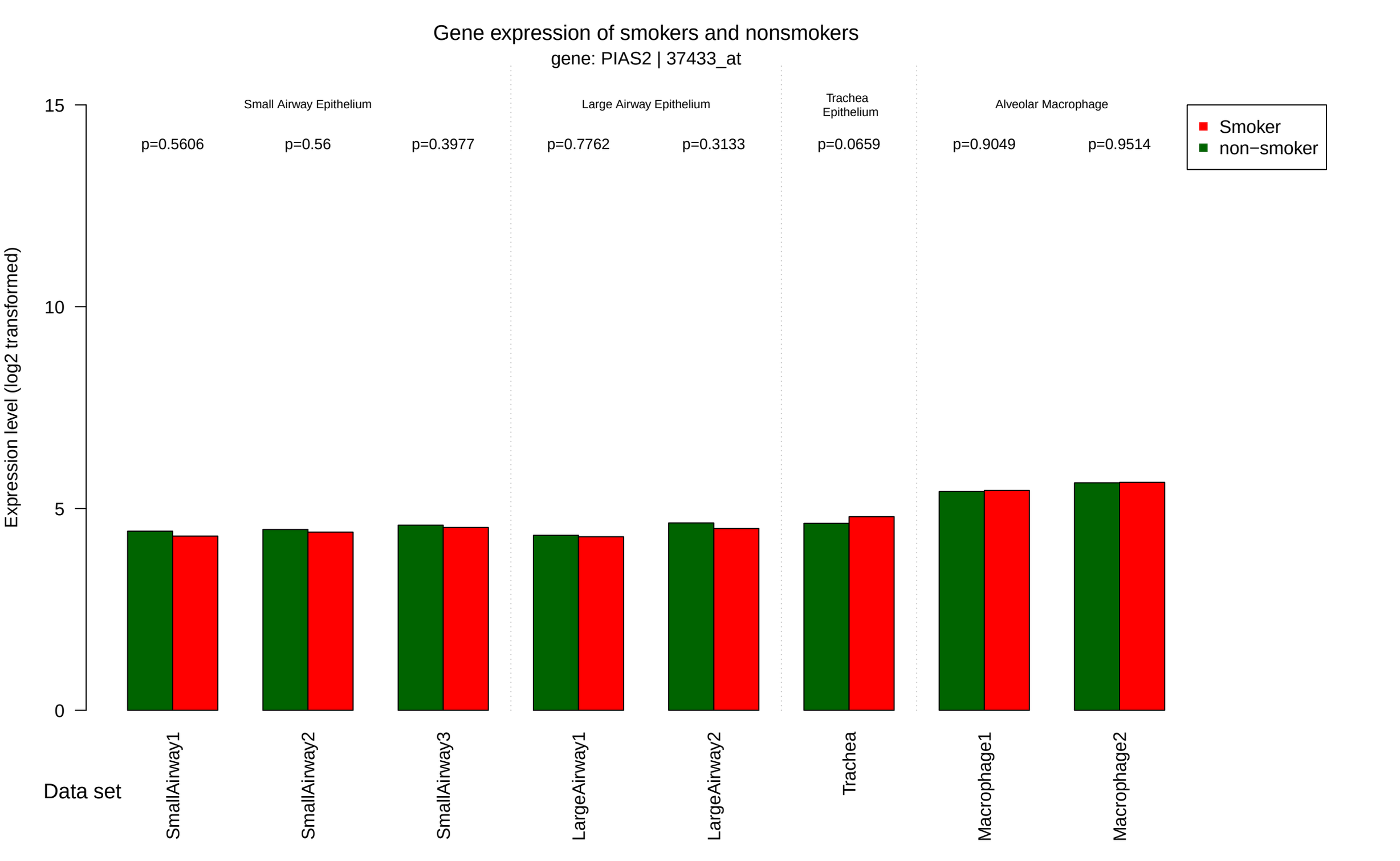
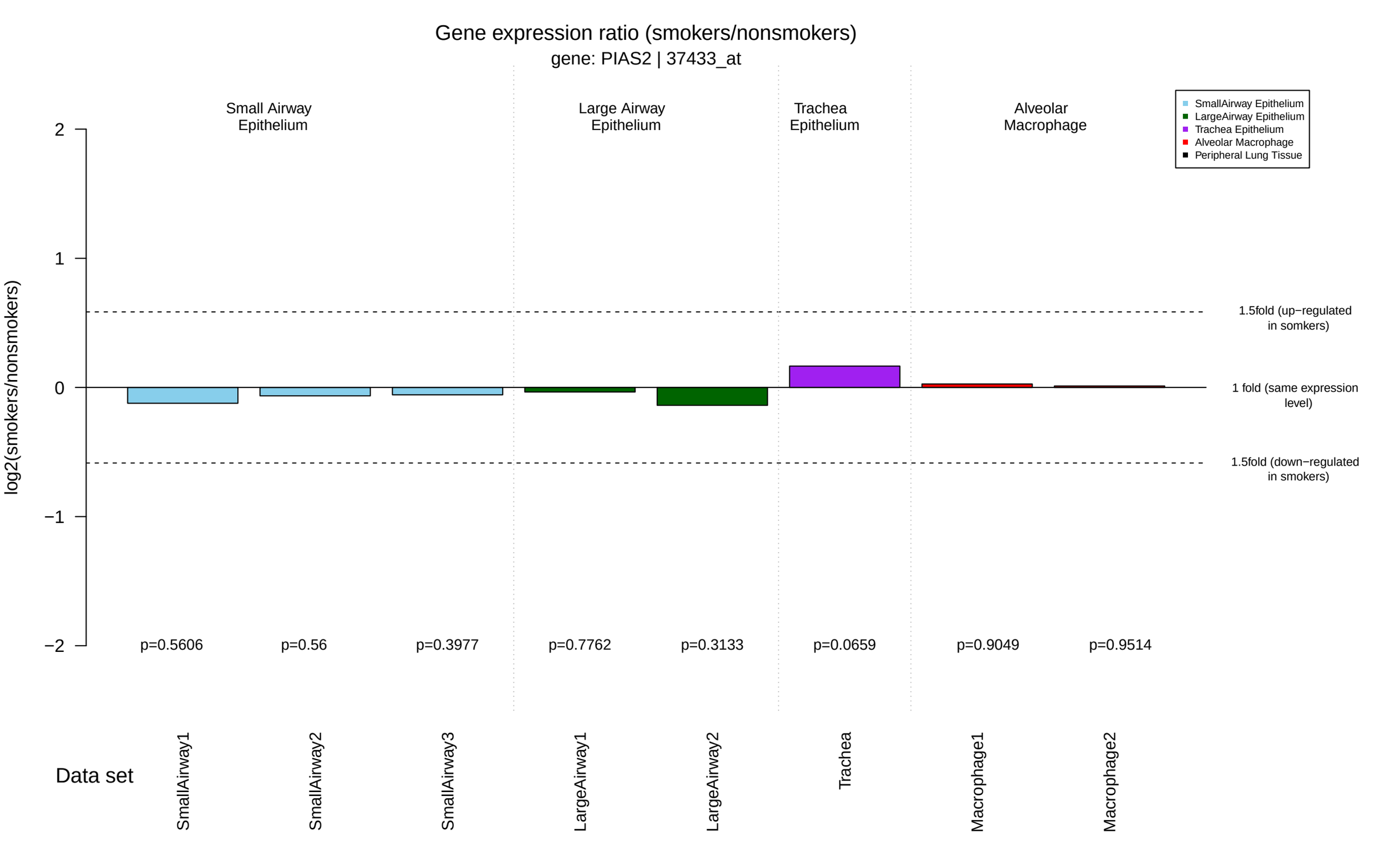
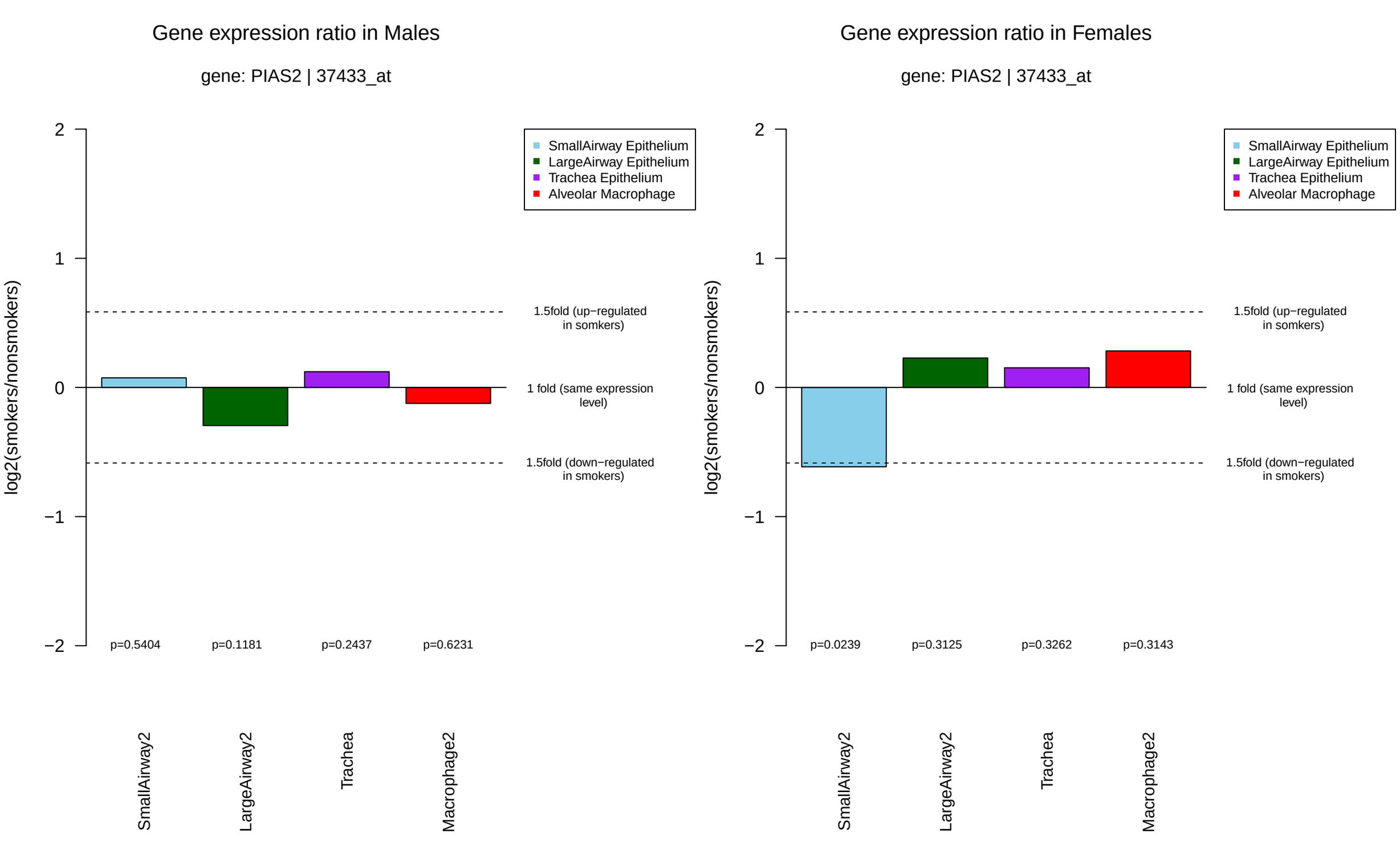
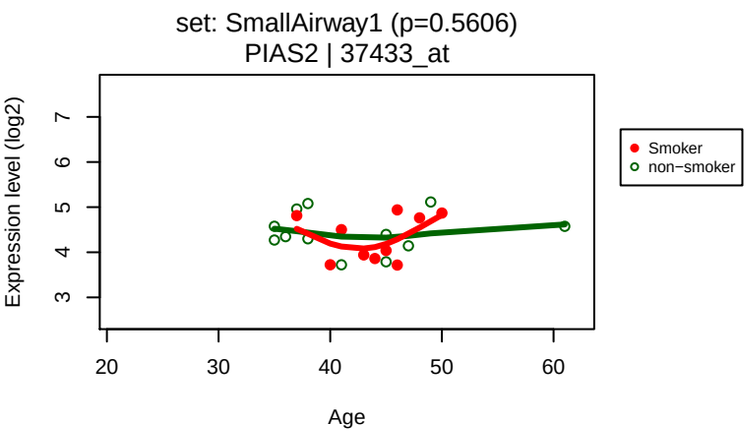
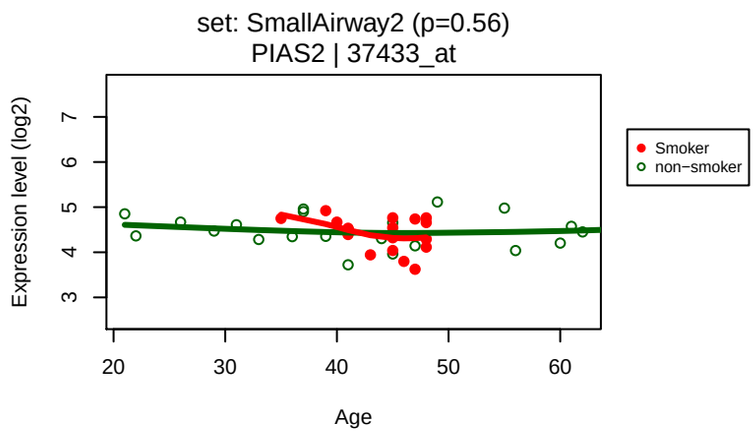
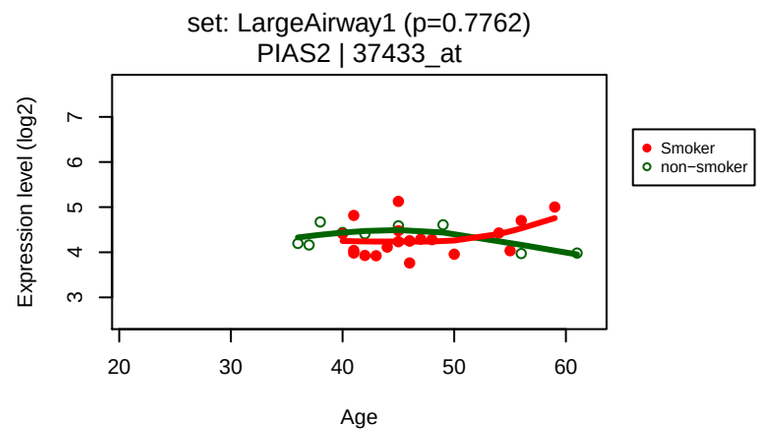

Gene expression of smokers and nonsmokers
gene: PIAS2 | 37433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5606
p=0.56
p=0.3977
p=0.7762
p=0.3133
p=0.0659
p=0.9049
p=0.9514
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

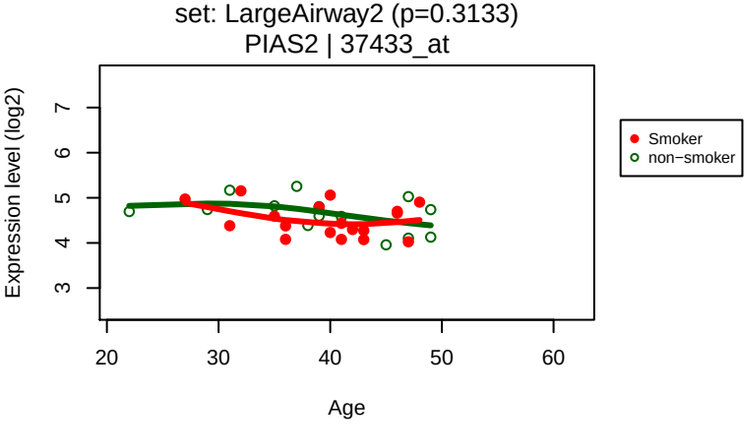

Gene expression ratio (smokers/nonsmokers)
gene: PIAS2 | 37433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5606
p=0.56
p=0.3977
p=0.7762
p=0.3133
p=0.0659
p=0.9049
p=0.9514
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

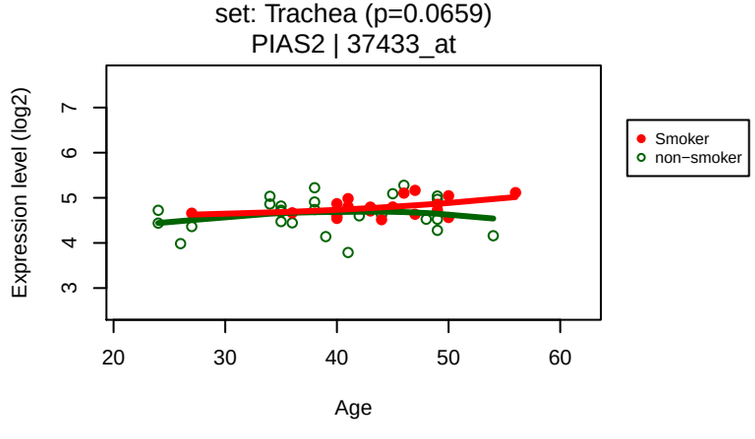

Gene expression ratio in Males
gene: PIAS2 | 37433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5404
p=0.1181
p=0.2437
p=0.6231
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS2 | 37433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0239
p=0.3125
p=0.3262
p=0.3143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

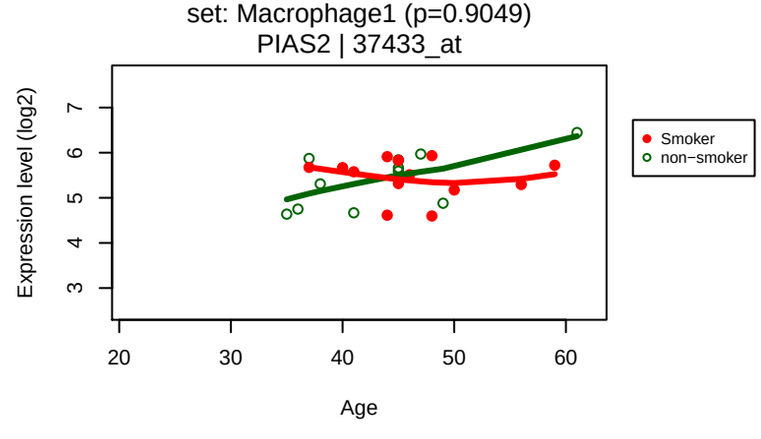

set: SmallAirway1 (p=0.5606)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker

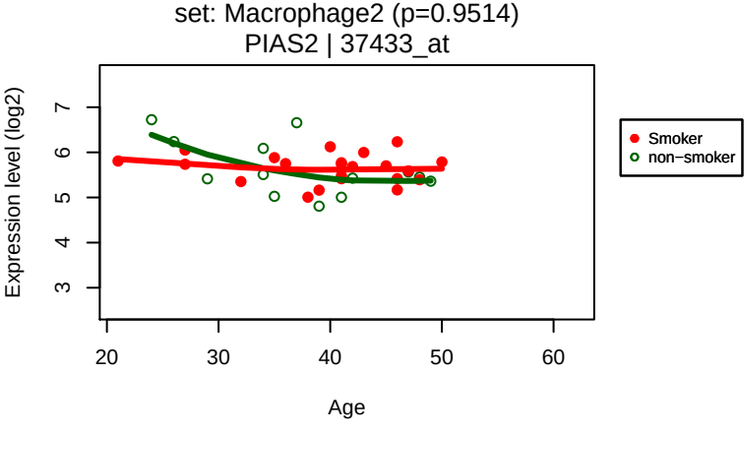

set: SmallAirway2 (p=0.56)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker

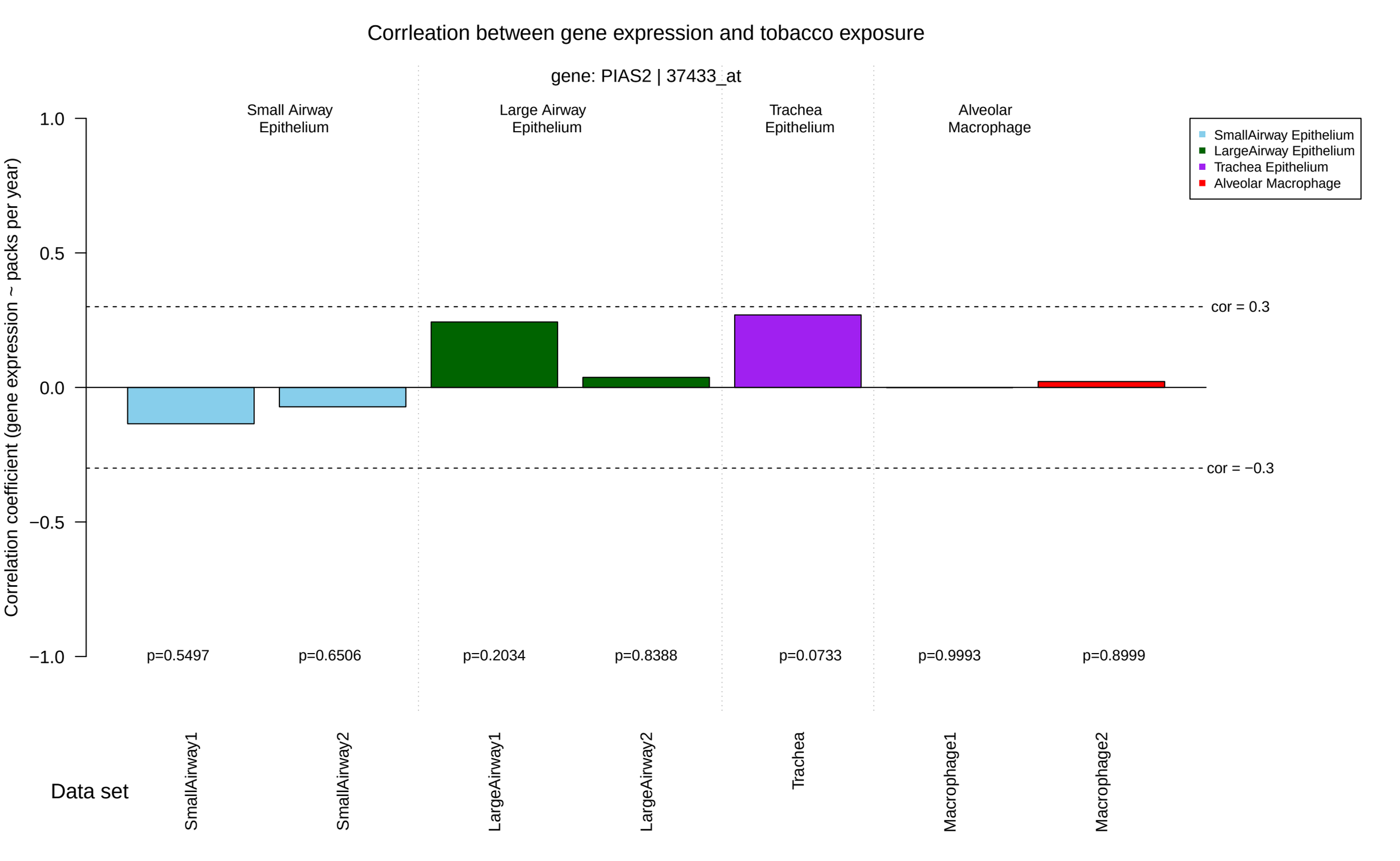

set: LargeAirway1 (p=0.7762)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3133)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0659)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9049)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9514)
PIAS2 | 37433_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS2 | 37433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5497
p=0.6506
p=0.2034
p=0.8388
p=0.0733
p=0.9993
p=0.8999
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2